Part I.

Project Descriptions

1. Mathematical Billiards

In this project, you will investigate the trajectories of a point particle in various billiard table geometries.

Project Instructions

For this project, use the additional resources available on Canvas.

Mathematical Billiards Simulation:

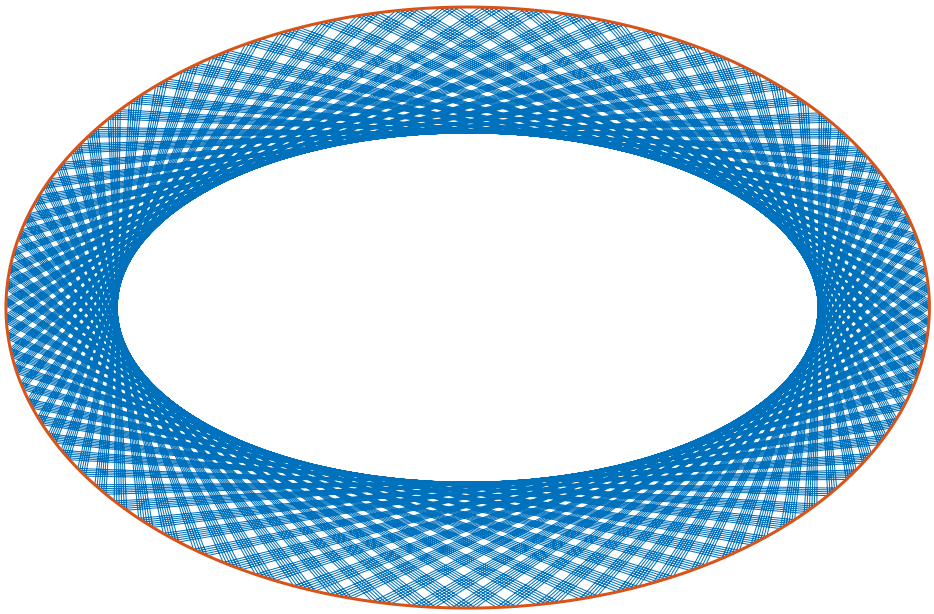
1. The billiards table is defined by a given geometry. The walls of the table are perfectly reflective.
2. The ball starts at an initial position (x_0, y_0) inside the table. The ball has an initial velocity (v_x, v_y).
3. The next reflection position can be calculated using just the current position and velocity.
4. When the ball hits a wall it experiences specular reflection where

$$\theta_{\text{incidence}} = \theta_{\text{reflection}}$$

Tasks

1. Simulate the trajectories of a point billiard ball in a box geometry (i.e., a rectangular table) where the ball reflects off the walls with equal angles of incidence and reflection.
2. Investigate and plot different trajectories for various initial conditions and velocities, highlighting any special or periodic trajectories.
3. Repeat your analysis for an elliptical geometry. Study how the elliptical boundary alters the motion of the ball. Are there any periodic orbits? Compare your results with the rectangular case.
4. Extend your investigation to the more complex geometry of stadium billiards, which consists of two semi-circular caps attached to a rectangular section. Simulate the trajectories, compare the system's behaviour to the box and elliptical geometries. Can you find any periodic orbits? By considering the Lyapunov exponent examine for evidence of chaotic motion in this geometry.

5. Visualise the phase space plots of all three systems. These plots should include parameters such as position, velocity, and angles of incidence at each bounce. Explore how varying initial conditions affects the phase space trajectories and determine whether the system shows signs of chaos, including sensitive dependence on initial conditions.



2. Forest Fire

In this project, you will explore models of forest fires to investigate the concept of self-organising criticality (SOC), where a system naturally evolves toward a critical state without external tuning.

Project Instructions

The Forest Fire Model:

In this project, the forest is represented as a grid where each cell can be in one of three states:

- Empty (0): A space without trees.
- Tree (1): A cell that contains a tree.
- Burning (2): A tree that is currently on fire.

The model evolves according to the following rules:

- Trees grow in empty cells with a probability p.
- Burning trees turn into empty cells in the next time step.
- A tree catches fire if one of its neighbouring cells is burning (nearest neighbours in the grid), or with a small probability f due to a spontaneous spark (lightning).

Tasks

1. One of your initial goals should be to produce a visualisation that shows a fire spreading to nearby trees in a grid. See images below. Animating this simulation is recommended for easier debugging. Write a program that implements the forest fire model on a grid. Set up parameters for tree growth rate (p) and the probability of a spontaneous fire (f). Run your simulation over multiple time steps and visualise the fire spreading through the forest. You will need to decide on a method to implement periodic boundary conditions.
2. Investigate the distribution of fire sizes. Specifically, determine if the distribution of fire events $N(s)$ follows a power law, which would indicate that the system is critical. Does the system self-organise towards criticality as you vary p and f?
3. Once the system reaches a critical regime, analyse the clusters of live trees that remain after a fire. Calculate the area and circumference of these clusters, and

investigate if other parameters can be used to describe the system. Use these observations to further characterise the system's critical behaviour.



Snapshots from an animation of the forest fire. Shows a single fire (Red) spreading to surrounding trees(Green) leaving behind an empty cell(Black).

H. J. Jensen, *Self-organised criticality: emergent complex behaviour in physical and biological systems*, Cambridge University Press, 1998

3. Monte Carlo Analysis of the Ising Model

In this project Monte Carlo simulations of ferromagnetism will be used to try to find a phase transition: The point at which a particular style of order in the system is lost.

Project Instructions

For this project, use the additional resources available on Canvas.

The square lattice ferromagnetic Ising model is the case in point:

$$H = \frac{J}{2} \sum_{<ij>} \sigma_i \sigma_j$$

with $< ij >$ denoting nearest-neighbours on the chosen geometry, the factor of two to compensate for double counting ($i \leftrightarrow j$) and $\sigma_i = \pm 1$ being a variable representing whether a particular spin is 'up' or 'down'.

Tasks

To get a feel for Monte Carlo methods we will begin with applying these methods to estimating π.

1. Write a program to make a statistical estimate for π using one of the two methods suggested in 'Maths13'.
2. Run your program many times and attempt to fit the distribution and assess errors.

We now proceed to using these methods to analyse the 1d Ising model.

1. Use the Metropolis algorithm and Monte Carlo to simulate the Ising model in 1d. For $T = 0.01J$ to $T = 4J$ calculate curves for:
   a) the internal energy per spin
   $$e(T) = \frac{\langle E \rangle}{N}$$
   b) the specific heat per spin
   $$C(T) = \frac{\langle E^2 \rangle - \langle E \rangle^2}{N k_B T^2}$$

2. Compare your results to the analytical ones found using transfer matrices.
3. Use your Monte Carlo to analyse the 2d ferromagnetic Ising model.
    a) Output the specific heat and magnetisation as a function of temperature, from $T = 0.01J$ to $T = 4J$.
    b) Investigate the phase transition, and attempt to find the correlation length and associated power law.

4. Satellites beyond Jupiter

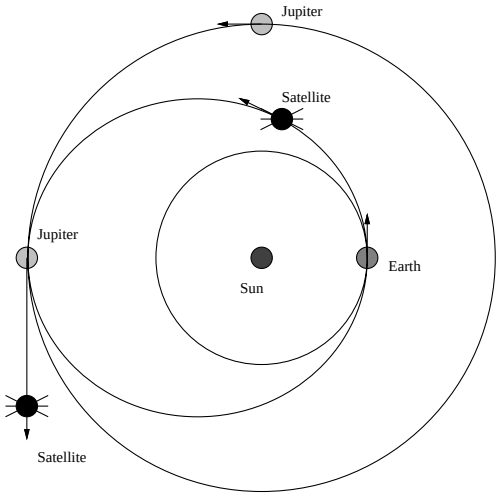
This project addresses the issue of how to use the existence of the planet Jupiter to save NASA money. A satellite is made on Earth and then needs to be transported to its eventual 'home' outside of the solar system. (c.f. the Voyager probes). There are two parts to this: one is called the "gravity assist" from Jupiter, and the second is the issue of how to transfer between orbits.

Project Instructions

For this project, use the additional resources available on Canvas.

At first sight, one might calculate the difference in potential energy between the initial and eventual positions of the satellite and then provide that in kinetic energy, at great expense to NASA. Although with only two bodies (the satellite and the Sun) there is no way to avoid the payment in potential energy, with three bodies (including Jupiter), the satellite can gain kinetic energy created from the potential energy between the other two. It is this effect that you will investigate: In simple terms, the satellite can pick up kinetic energy by 'bouncing off' Jupiter, known as a "gravity assist".

In order to transfer between the Earth's orbit and Jupiter's, one uses a *transfer orbit*. An example of this is the Hohmann transfer orbit, as depicted below:



In order to minimise the expense, the first task is to decide *when* to launch in order to

achieve close approach to Jupiter with the minimum orbit depicted. The next task is to find out the best configuration for the satellite and Jupiter when they meet, in order to achieve the maximum impulse to the satellite.

Tasks

1. **Transfer Orbit:**
   a) Calculate *when* to launch the satellite, based upon the radii of the orbits of Earth and Jupiter, assuming that both planets have *circular* orbits and ignoring *all* potentials except that of the sun.
   b) Calculate the required impulse for this transfer.
   c) Generate a system of first order equations for the motion of the satellite and verify the transfer orbit numerically.
2. **Gravity Assist:**
   a) Extend your calculation to incorporate Jupiter's gravitational potential, and decide a strategy to gain from the motion of Jupiter.
   b) Can the satellite escape from the Solar system?
   c) Can you escape the solar system using Mars? (You will need to look up Mars data).
3. **Data:**

Gravitational Constant	$0.667 \times 10^{-10} m^3 kg^{-1} s^{-2}$
Mass of the Sun =	$0.1984 \times 10^{31} kg$
Mass of the Earth =	$0.5976 \times 10^{25} kg$
Mass of Jupiter =	$0.1903 \times 10^{28} kg$
Radius of the Earth's Orbit =	$0.1495 \times 10^{12} m$
Radius of Jupiter's Orbit =	$0.7778 \times 10^{12} m$
Radius of the Earth =	$0.6368 \times 10^{7} m$
Radius of Jupiter =	$0.6985 \times 10^{8} m$

5. Random Walks

In this project, you will investigate certain properties of random walks.

Project Instructions

A random walk consists of a connected path formed by randomly adding new bonds to the end of the existing walk.

Tasks

1. **Numerical**:
   a) Write a program to create a random walk on a square lattice of given length.
   b) Visualise the random walk.
   c) Evaluate the distances between all pairs of points at a fixed number of steps apart along your walk.
   d) Try to fit your distribution of distances and find the best value for the variance.
   e) Assess the assertion that the variance scales as N for N steps.
   f) Repeat these simulations for a triangular lattice, and decide whether the variance also scales as N.
2. **Investigations**:
   a) For your chosen lattice, write a program to find a self-avoiding random walk.
   b) It is often asserted that the variance of a self-avoiding random walk scales with a different power of N. What do you think of this assertion?

6. Snakes and Ladders

In this project, you'll investigate the board game Snakes and Ladders to explore random processes in games.

Project Instructions

For this project, use the additional resources available on Canvas.

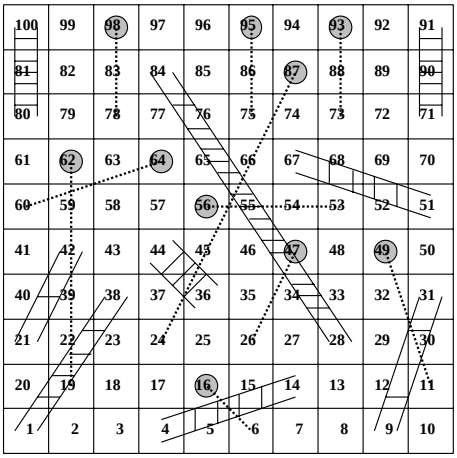
Snakes and Ladders:

- In this version of snakes and ladders there is a single player.
- The game is played on a 10×10 grid, numbered 1 to 100.
- Certain squares have ladders that advance the player upwards (to a higher-numbered square), and others have snakes that send the player downwards (to a lower-numbered square).
- The player starts at position 0. You don't need to add position 0 to your visualisation but is important for the later Markov Chain theory.
- The objective is to reach *or pass* square 100.
- The player rolls a die (with values $1 - 6$) to determine how many spaces to move forward.
- The player moves forward by the number of spaces rolled.
  - If the player lands on the base of a ladder, they immediately move up to the top of the ladder.
  - If the player lands on the head of a snake, they immediately move down to the tail of the snake.
- In this version of the game the player's move must take them to or past 100 to win. This is a slight variation on the standard rules which are to land exactly on 100 to win.

Tasks

1. Write a program to simulate a game of Snakes and Ladders based on the provided game board. Your simulation should handle the movement of a single player according to the rules above.

2. Run multiple iterations of the game to collect data on the duration of each game session (i.e., the number of dice rolls taken to reach the end).
3. Calculate the average duration of the game over all simulations and assess the variability in game lengths by computing the variance of the duration. Present this data clearly, using appropriate statistical methods and visualisations.
4. Apply the theory of absorbing Markov Chains to analyse the game dynamics. Define the states corresponding to the squares on the board, considering the snakes and ladders as transitions between states.
5. Use Markov Chain theory to calculate the expected duration of the game (i.e., the expected number of rolls to finish) and its variance. Compare these analytical results with your numerical simulation findings to evaluate the accuracy and reliability of your simulation model.



7. Strange Attractor

In this project, you will investigate the behaviour of a forced, damped harmonic oscillator, a non-integrable, dissipative dynamical system.

Project Instructions

For this project, use the additional resources available on Canvas.

This project focuses on the numerical investigation of a *non-integrable, dissipative dynamical system* – a forced, damped, harmonic oscillator, governed by the following equation:

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \nu\frac{\mathrm{d}\theta}{\mathrm{d}t} + \sin\theta = T\sin(2\pi f t)$$

where:

- f is the forcing frequency,
- T is the forcing amplitude,
- ν is the friction coefficient.

Tasks

1. Write a program to numerically integrate the system's equation. A good integrator for the problem is `integrate.odeint` in the scipy library. Simulate the system's motion for a variety of initial conditions and parameter values (e.g., ν, T, and f). You may find it helpful to investigate simpler systems as a starting point to verify your numerical approach. A few examples of simpler systems are:
   a) The undamped, unforced harmonic oscillator: $\ddot{\theta} + \theta = 0$,
   b) The damped harmonic oscillator: $\ddot{\theta} + \nu\dot{\theta} + \theta = 0$,
   c) The forced harmonic oscillator (without $\sin\theta$): $\ddot{\theta} + \nu\dot{\theta} + \theta = T\sin(2\pi f t)$.
2. Once the code is verified, generate a *Poincaré section* by plotting the system's state:
$$\left(\theta(t_n), \frac{\mathrm{d}}{\mathrm{d}t}\theta(t_n)\right)$$
   at times $t_n = \frac{n}{f}$ (i.e., at the same point in each forcing cycle). Wait for transients to decay, and plot the steady-state attractor.

3. Explore the chaotic behaviour by increasing the forcing amplitude T while keeping ν fixed in the *not-overdamped* regime. Analyse how the system's behaviour changes and whether the attractor becomes non-trivial or chaotic.
4. Use the *box-counting method* to estimate the dimension of the attractor from your Poincaré section. Determine whether the attractor exhibits fractal properties, indicative of a *strange attractor*.
5. Examine how the attractor evolves as parameters like T and ν are adjusted. Identify regions where the attractor's structure changes or becomes chaotic.

8. Topological Insulators

In this project, you will investigate the Su-Schrieffer-Heeger (SSH) model, a fundamental 1D tight-binding model that demonstrates topological phase transitions and zero-energy protected edge states.

Project Instructions

The SSH Model Hamiltonian:

In a 1D bipartite lattice with sublattices A and B, the Hamiltonian is characterized by alternating intracell (v) and intercell (w) hopping amplitudes.

$$\begin{aligned} H = &\sum_{j=1}^{N} v|j,A\rangle\langle j,B| + v|j,B\rangle\langle j,A| \\ &+ \sum_{j=1}^{N-1} w|j+1,A\rangle\langle j,B| + w|j,B\rangle\langle j+1,A| \end{aligned} \tag{8.1}$$

Tasks

1. Construct the $2N \times 2N$ real-space Hamiltonian matrix H_{OBC} under open boundary conditions. Numerically diagonalize the matrix for $N = 50$ unit cells across a range of dimerization parameters $\delta \in [-0.8, 0.8]$, where $v = 1 - \delta$ and $w = 1 + \delta$.
2. Plot the real-space energy spectrum against δ. Demonstrate the emergence of two zero-energy ($E \approx 0$) mid-gap states in the topological regime ($\delta > 0$) and plot the probability density $|\psi(x)|^2$ of these eigenvectors to demonstrate spatial localization at the boundaries.
3. By projecting the zero-energy eigenvalue equation $H|\psi\rangle = 0$ onto each sublattice site, $\langle j,B|H|\psi\rangle = 0$ and $\langle j,A|H|\psi\rangle = 0$, derive the recurrence relations for the wavefunctions on the A and B sublattices. Use these to find the closed-form analytical expression for the left- and right-edge wavefunctions, and overlay these analytical solutions on your numerical eigenvectors from the previous task.
4. Introduce off-diagonal (hopping) disorder $v_m = v + \delta v_m$ and diagonal (onsite) disorder ϵ_m. Investigate how chiral symmetry protects the zero-energy edge states

against hopping disorder, and demonstrate how onsite disorder breaks chiral symmetry to lift the zero-mode degeneracy.

5. Simulate the time evolution of a state initially localized at the edge, $|\psi(0)\rangle = (1, 0, \dots)^T$, under the time evolution operator $U(t) = e^{-iH_{\text{OBC}}t/\hbar}$. Compare the survival probability at the boundary $P_{\text{edge}}(t) = |\langle\psi(0)|\psi(t)\rangle|^2$ for trivial vs. topological regimes following a quantum quench.

References:

J. K. Asbóth, L. Oroszlány, and A. Pályi, *A Short Course on Topological Insulators: Band Structure and Edge States in One and Two Dimensions*, Lecture Notes in Physics, Springer (2016).